

GEO Satellite Communication Systems

How Does Communication Theory Explain the Performance of This Real System?

LEEN354 – Communication Theory II | Istanbul Arel University | April 2026

Hüseyin Kaya | 220303036

Instructor: Assist. Prof. Dr. Semra KOCAASLAN KARAMZADEH

What is a GEO Satellite?

- Geostationary Earth Orbit: altitude = 35,786 km
- Orbital period = 24 hours → appears stationary from ground
- Single satellite covers $> 120^\circ$ longitude (1/3 of Earth)
- One-way propagation delay ≈ 270 ms
- Free-space path loss > 200 dB — the central engineering challenge

Communication theory answers:

How does the signal survive 200+ dB of loss?

Reference Systems

Türksat 4A

42°E, Ku-band, Turkey & region

Intelsat IS-901

C/Ku-band, global fleet rep.

GEO Satellite Link Architecture

Figure 1 — GEO Satellite Communication System Block Diagram



Three segments: Uplink (TX → Satellite) | Transponder (frequency translation + HPA) | Downlink (Satellite → RX)

AWGN thermal noise injected at LNA stage on receive side

Communication Theory Toolbox

Link & Channel

$$\text{FSPL [dB]} = 20 \cdot \log_{10}(4\pi df/c)$$

Free-space path loss

$$\text{C/N} = \text{EIRP} + \text{G/T} - \text{FSPL} - 10 \cdot \log_{10}(\text{kB}) - \text{L}$$

Link budget equation

$$r(t) = s(t) + n(t), \quad n \sim N(0, \sigma^2)$$

AWGN channel model

Performance Limits

$$P_e(\text{BPSK}) = Q(\sqrt{2 \cdot E_b/N_0})$$

Bit Error Rate — BPSK/QPSK

$$C = B \cdot \log_2(1 + \text{SNR})$$

Shannon channel capacity

$$\gamma_r = k \cdot R^a \rightarrow A = \gamma_r \cdot d_e^2$$

ITU-R P.838 rain attenuation

These 6 equations completely explain GEO satellite performance

Reference System — Türksat 4A (Ku-Band)

Parameter	Value	Unit
GEO Altitude	35,786	km
Downlink Frequency (Ku)	12	GHz
Satellite EIRP	52	dBW
Earth Station G/T	20	dB/K
Transponder Bandwidth	36	MHz
System Noise Temperature	150	K
Total Miscellaneous Loss	1.3	dB

Simulation Tools

Python 3

NumPy · SciPy · Matplotlib

Monte Carlo: 200,000 bits / point

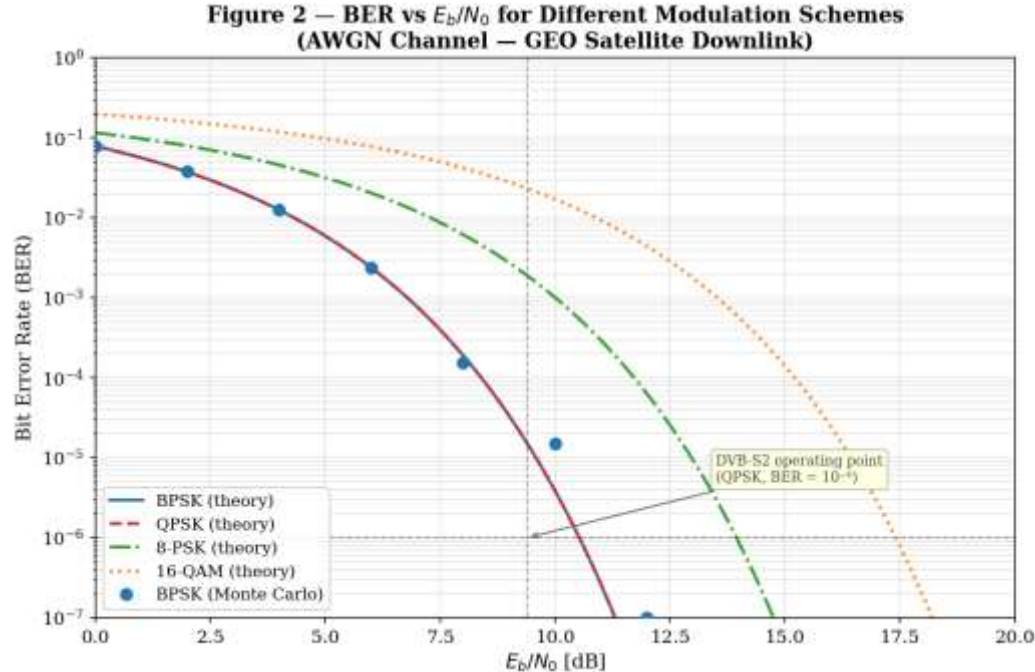
Key Result

C/N = 18.64 dB

(Ku-band downlink)

Türksat 4A ref. mission

Bit Error Rate vs E_b/N_0 — Modulation Comparison



BPSK = QPSK in BER, but QPSK is 2× more spectrally efficient

BER = 10^{-6} needs:

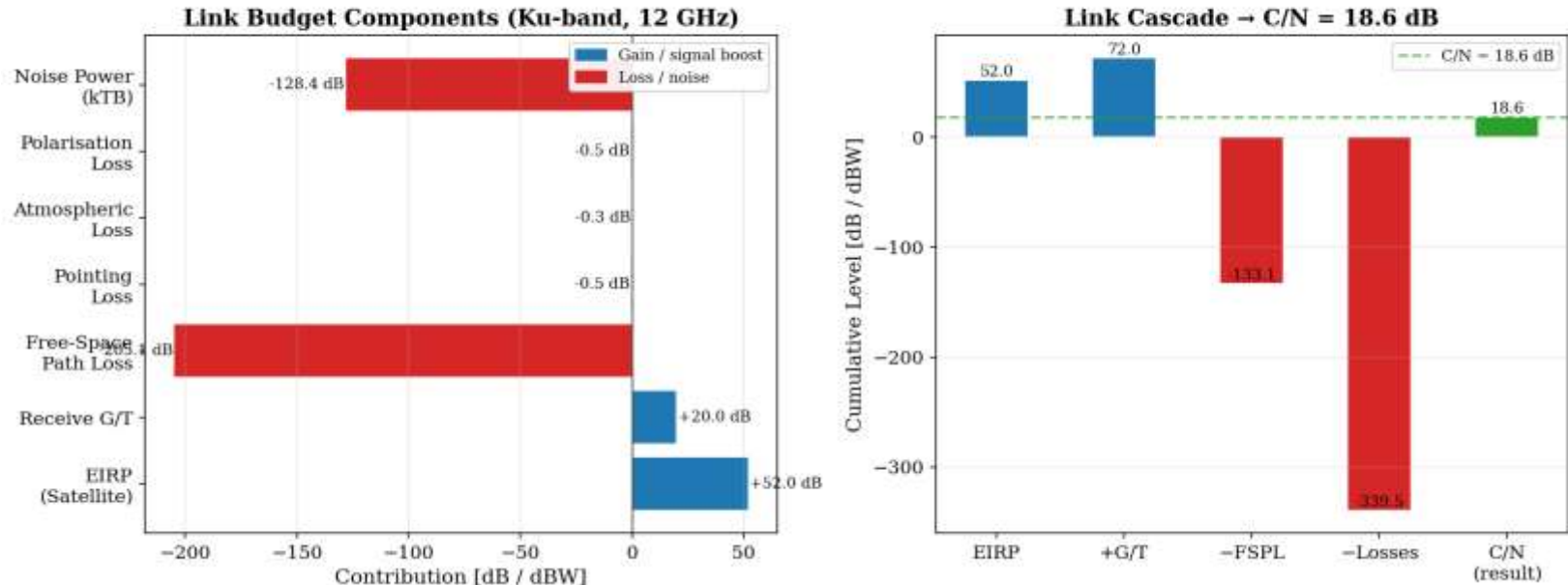
QPSK: 10.5 dB | 8-PSK: 14 dB | 16-QAM: 17.5 dB

Monte Carlo (circles) confirms theory within $\pm 5\%$

DVB-S2 op. point annotated at 9.4 dB

Ku-Band Link Budget — Step by Step

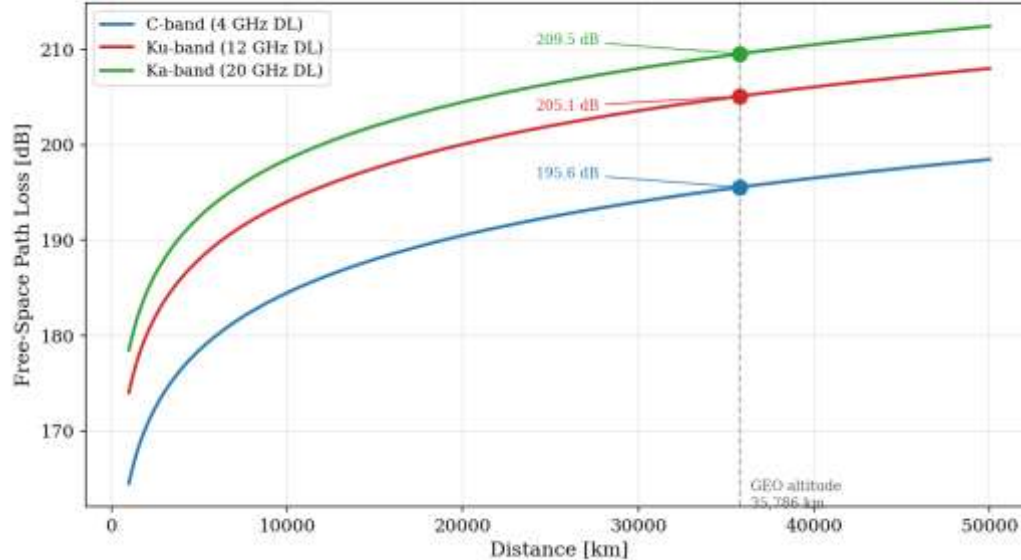
Figure 3 — Ku-Band GEO Satellite Link Budget Analysis



$C/N = 52 + 20 - 205.10 - 1.3 + 153.04 = 18.64 \text{ dB} \checkmark$ | Comfortable margin above DVB-S2 QPSK threshold

Free-Space Path Loss — Why 200+ dB?

Figure 4 — Free-Space Path Loss vs Distance for C, Ku, and Ka-band GEO Links



C-band (4 GHz)

195.56 dB

Baseline

Ku-band (12 GHz)

205.10 dB

+9.5 dB

Ka-band (20 GHz)

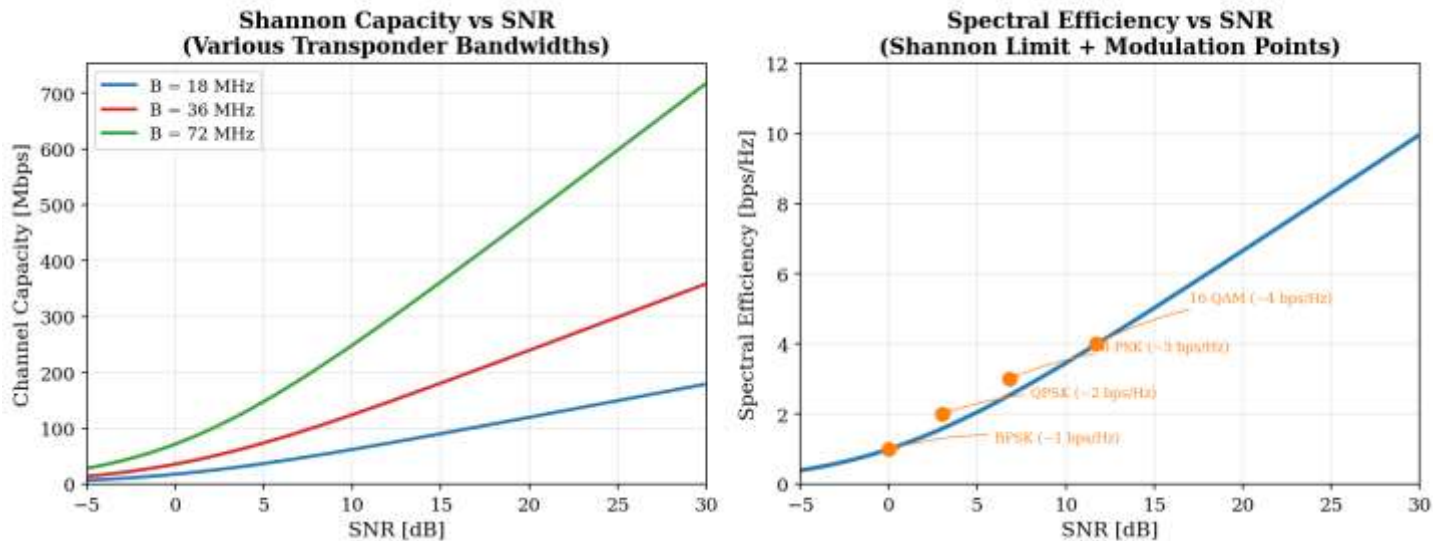
209.54 dB

+14.0 dB

Higher frequency → higher path loss → smaller antennas possible, but more transmit power required

Shannon Limit — How Much Can We Transmit?

Figure 5 — Shannon Channel Capacity for GEO Satellite Links



DVB-S2 QPSK @ ~2 bps/Hz → gap to Shannon limit filled by ACM + LDPC coding

223.6 Mbps

Shannon limit
(36 MHz, C/N=18.64 dB)

6.21 bps/Hz

Max spectral efficiency

~2 bps/Hz

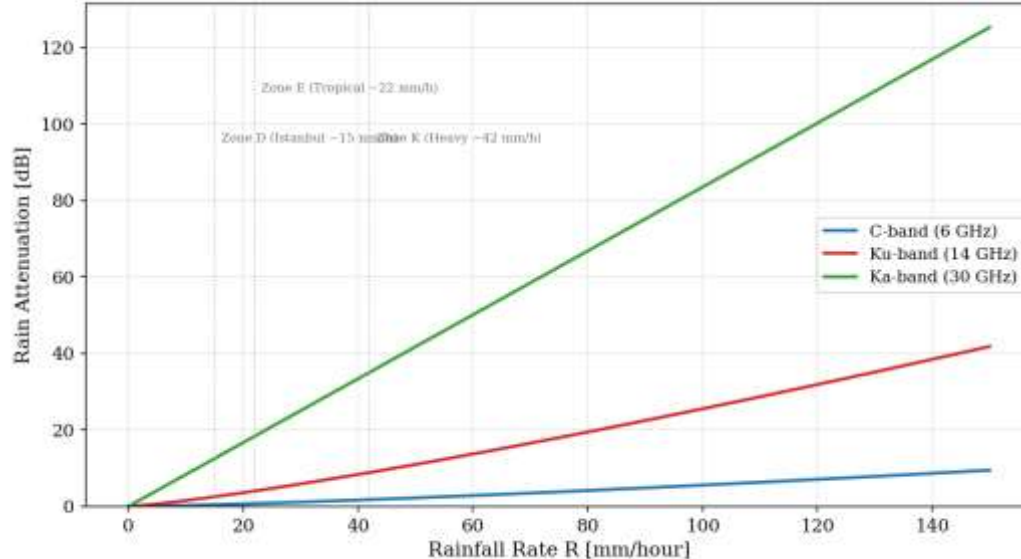
DVB-S2 QPSK (practical)

~5 bps/Hz

DVB-S2X 32-APSK (best)

Rain Attenuation — Why Frequency Choice Matters

Figure 6 — Rain Attenuation vs Rainfall Rate
(ITU-R P.838 Model, Path Length = 5 km, Uplink Frequencies)



Istanbul | R = 42 mm/h (Zone K)

C-band (6 GHz)

≈ **1.0 dB**

Negligible ✓

Ku-band (14 GHz)

≈ **7.5 dB**

Manageable

Ka-band (30 GHz)

≈ **37 dB**

CATASTROPHIC

ITU-R P.838 | $\gamma_r = k \cdot R^\alpha$ → power law causes 37× difference between C-band and Ka-band

Simulation Results at a Glance

Analysis	Result	Theory Used	Match?
FSPL (Ku-band, 12 GHz, 35,786 km)	205.10 dB	Eq. 1 — FSPL	✓
Link Carrier-to-Noise Ratio	18.64 dB	Eq. 2 — Link Budget	✓
QPSK BER = 10^{-6} threshold	$E_b/N_0 = 10.5$ dB	Eq. 4 — QPSK BER	✓
Shannon Capacity (36 MHz transp.)	223.6 Mbps	Eq. 7 — Shannon	✓
Rain att. Ka-band (42 mm/h, 5 km)	≈ 37 dB	Eq. 8-9 — ITU-R P.838	✓
Monte Carlo vs. Analytical BER	± 5 % agreement	Statistical validation	✓

Trade-Offs Revealed by Communication Theory

Modulation

- $\uparrow$ order $\rightarrow \uparrow$ efficiency BUT $\uparrow E^b/N_0$ needed
- QPSK: robustness + efficiency balance
- ACM: dynamically switches during rain events
- 7 dB penalty from QPSK to 16-QAM = 7 dB less rain margin

Frequency Band

- C-band: rain-robust, large antennas, limited spectrum
- Ku-band: compact 0.6 m terminals, moderate rain risk
- Ka-band: 2.5 GHz+ spectrum, but severe rain fade
- HTS spot beams overcome Ka path loss penalty

Link Budget

- 1 dB FSPL increase $\rightarrow$ 1 dB C/N decrease (direct mapping)
- G/T: 2.4 $\rightarrow$ 4.5 m dish gives +6 dB margin
- Ka rain at 30 mm/h: 25 dB fade $\rightarrow$ site diversity needed
- Equipment aging over 15-year life needs margin buffer

Future Extensions

1

DVB-S2 LDPC+BCH coding simulation

Quantify ~9 dB coding gain → close the gap to Shannon limit

2

ACM with real Istanbul rainfall data

Link availability: 99.5% (fixed QPSK) → 99.9% (ACM)

3

LEO vs GEO quantitative comparison

Starlink: 550 km altitude, 2 ms delay, FSPL $\approx$ 170 dB

4

HTS multi-beam frequency reuse

Fleet capacity 100–500 Gbps (Eutelsat KONNECT VHTS)

5

Nonlinear HPA modelling (Saleh)

AM/AM distortion → spectral regrowth & interference

Conclusion

Communication theory COMPLETELY explains GEO satellite performance:

205 dB path loss

Link budget equation (Eq. 2) dictates every single design choice

QPSK = optimal

Identical BER to BPSK, 2× spectral efficiency — DVB-S2 baseline modulation

Shannon: 223.6 Mbps

ACM + LDPC coding continuously narrows the gap toward the theoretical limit

Ka rain fade: 37 dB

Quantitative proof that site diversity is not optional — it is mandatory

Thank you for your attention